ABDUL JANEEB K

ADJUST TUNING PROFILES

- Tuning is a customization to improve system perfomace
- we can use two method for tuning
- static tuning
- dynamic tuning
- here we focusing about static tuning
- static tuning refers to configuring system performance settings using the tuned service and tuning profiles
- The benefit of tuning in Linux is to improve system performance according to the system's purpose.
- For example:
- balanced — balances performance and power consumption.
- throughput-performance — optimizes for high data throughput.
- powersave — aims to reduce power consumption.
- Simple example: If a Linux server handles many requests, you can select a profile suited to high throughput.
- Yum install -y tuned
- To check which profile is active #tuned-adm active
- Display how many profiles is available #tuned-adm list.
- Change a profile to another one #tuned-adm profile powersave
- If we need to see recommended profile #tuned-adm recommend

Important Tuned Profiles in Linux (RHCSA)

1. balanced

- **Purpose:** Balances system performance and power consumption.
- **Use:** General everyday workloads.

2. throughput-performance

- **Purpose:** Optimizes the system for high throughput.
- **Use:** Servers handling heavy workloads and large amounts of data.

3. powersave

- **Purpose:** Prioritizes reducing power consumption.
- **Use:** When power saving is important.

4. virtual-guest

- **Purpose:** Optimizes settings for a virtual machine.
- **Use:** Linux running inside VMware or another hypervisor.

Important Commands

List available profiles:

tuned-adm list

Check the active profile:

tuned-adm active

Check the recommended profile:

tuned-adm recommend

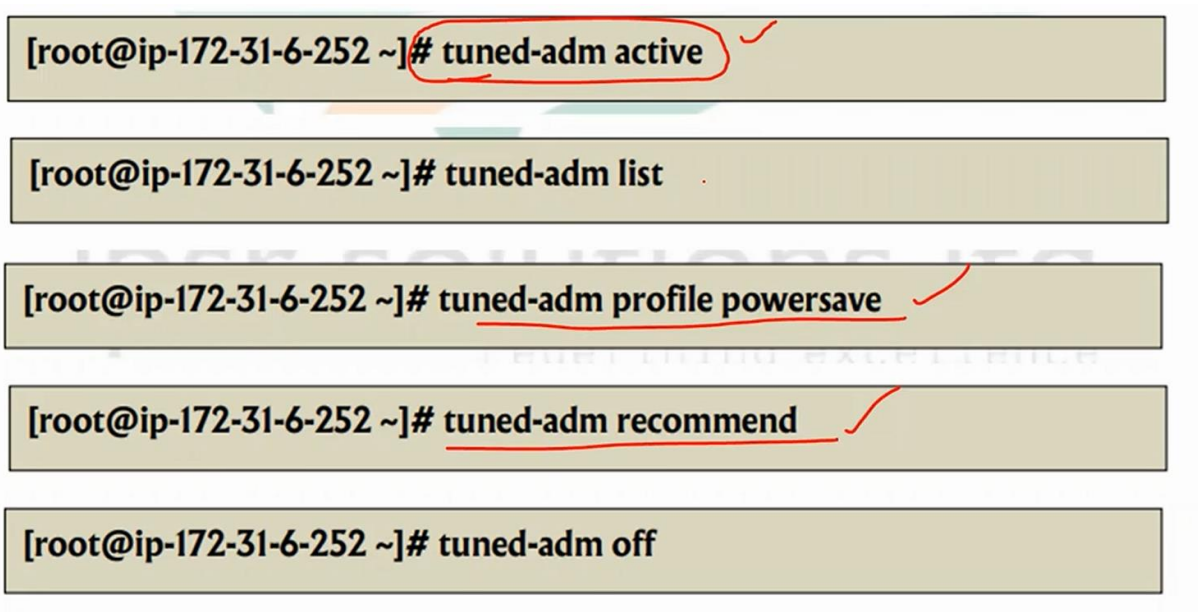
Apply a profile:

tuned-adm profile balanced

Remember for RHCSA:

- balanced — General use
- throughput-performance — High throughput
- powersave — Power saving
- virtual-guest — Virtual machines

screenshots



```
login as: root
root@192.168.1.158's password:
Last login: Sat Oct 17 09:30:30 2026 from 192.168.1.5
[root@Astra ~]# yum install -y tuned
Last metadata expiration check: 0:39:11 ago on Sat 17 Oct 2026 10:21:37 AM IST.
Package tuned-2.27.0-2.el9_8.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@Astra ~]# systemctl status tuned
● tuned.service - Dynamic System Tuning Daemon
     Loaded: loaded (/usr/lib/systemd/system/tuned.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-17 10:37:17 IST; 23min ago
       Docs: man:tuned(8)
             man:tuned.conf(5)
             man:tuned-adm(8)
   Main PID: 4721 (tuned)
      Tasks: 4 (limit: 7621)
     Memory: 14.2M (peak: 17.1M)
        CPU: 1.414s
     CGroup: /system.slice/tuned.service
             └─4721 /usr/bin/python3 -Es /usr/sbin/tuned -l -P

Oct 17 10:37:16 Astra systemd[1]: Starting Dynamic System Tuning Daemon...
Oct 17 10:37:17 Astra systemd[1]: Started Dynamic System Tuning Daemon.
```

```
[root@Astra ~]# tuned-adm list
Available profiles:
- accelerator-performance     - Throughput performance based tuning with disabled higher latency STOP states
- aws                         - Optimize for aws ec2 instances
- balanced                    - General non-specialized tuned profile
- balanced-battery            - Balanced profile biased towards power savings changes for battery
- desktop                     - Optimize for the desktop use-case
- hpc-compute                 - Optimize for HPC compute workloads
- intel-sst                   - Configure for Intel Speed Select Base Frequency
- latency-performance         - Optimize for deterministic performance at the cost of increased power consumption
- network-latency             - Optimize for deterministic performance at the cost of increased power consumption, focused on low latency network performance
- network-throughput          - Optimize for streaming network throughput, generally only necessary on older CPUs or 40G+ networks
- optimize-serial-console     - Optimize for serial console use.
- powersave                   - Optimize for low power consumption
- throughput-performance      - Broadly applicable tuning that provides excellent performance across a variety of common server workloads
- virtual-guest               - Optimize for running inside a virtual guest
- virtual-host                - Optimize for running KVM guests
Current active profile: virtual-guest
[root@Astra ~]# tuned-adm active
Current active profile: virtual-guest
[root@Astra ~]# tuned-adm off
[root@Astra ~]# tuned-adm recommend
virtual-guest
[root@Astra ~]# tuned-adm profile virtual-guest
[root@Astra ~]# tuned-adm active
Current active profile: virtual-guest
[root@Astra ~]#
```